

AutoXR: Real-Time AR/VR Vehicle Customization

Final Project Initial Memo

Group 5: Vinay Vuyyuri, Mohammed Tanvir Rasheed, Ramana Prabhu Sana, Pallavi Yerramilli

Product Details: AutoXR



A software engine that, with a compatible XR headset, projects high-fidelity 3D renders of paints and parts onto the real vehicle, aligning to the car and its surroundings for true, life-size customization.

Abstract:

Choosing paints and parts is visual and costly, but current tools don't show true color or fit the actual car. AutoXR lets buyers and sales teams see realistic, full-scale options on the real vehicle and then send a single, priced package to a dealer or shop. The result is faster decisions, fewer mistakes, and clearer upsells.

Methodology:

Study plan: Configurations, confidence to buy, and conversion to quote.

Analysis:

- Screen related use cases;
- Build an EVC model for buyers/dealers/shops;
- Run a short survey/conjoint for pricing;
- A/B test AR vs. non-AR preview and lighting realism with guardrails (latency, frame rate, comfort);
- Check competitive risks and data/partner advantages.

Applications:

- Showrooms: quick demos that raise attachment to options.
- At home: view your own car and request a priced bundle.
- Custom shops: reduce returns and fitment disputes.
- OEM/brands: limited-edition trims and sponsored part libraries.

Market:

Starting with U.S. dealers and independent shops that sell high-option vehicles and accessories. Monetize via:

- Dealer/Shop console: subscription for catalog upload, simple price rules, and basic analytics.
- Brands/OEMs: license asset packs. Size the market from option/accessory spend, then set good/better/best pricing using conjoint and EVC ceilings (time saved, higher attach rate, fewer reworks). Launch with 3–5 pilots, build case studies, and partner with wheel/paint brands to seed trusted catalogs.